

First Chapter Title

Your Name

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1 Introduction

This chapter fixes notation and recalls standard definitions from modal logic.

2 The Modal Language

We work in a modal language $\mathcal{L}^{\mathbf{M}}$ with variables $\mathbf{Prop} = \{p, q, r, \dots\}$, the Boolean connectives, and a unary operator $\Box$ (necessity). The dual $\Diamond A := \neg\Box\neg A$ is read “possibly A .”

Definition 2.1. A *Kripke frame* is a pair $\mathfrak{F}_R = \langle W, R \rangle$ where $W \neq \emptyset$ and $R \subseteq W \times W$.

Definition 2.2. A *Kripke model* $\mathfrak{M}_R = \langle W, R, V \rangle$ adds a valuation $V : \mathbf{Prop} \rightarrow \wp(W)$ to a frame. The modal clause is:

$$\mathfrak{M}_R, w \Vdash \Box A \text{ if and only if for all } v, wRv \text{ implies } \mathfrak{M}_R, v \Vdash A.$$

3 Normal Modal Logics

The smallest normal modal logic, **K**, extends classical propositional logic with:

$$\begin{array}{ll} \mathbf{K} & \Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \\ \text{Nec} & \text{From } \vdash A, \text{ infer } \vdash \Box A. \end{array}$$

Familiar extensions include **KT** = **K** $\oplus$ **T** (where **T** is $\Box A \rightarrow A$) and **S4** = **KT** $\oplus$ ($\Box A \rightarrow \Box\Box A$).

Theorem 3.1. **K** is sound and complete with respect to the class of all Kripke frames [see 1, ch. 4].

Proof. Soundness: the axiom **K** is valid in every frame and Nec preserves validity. Completeness: construct the canonical model $\mathfrak{M}^c = \langle W^c, R^c, V^c \rangle$ where W^c is the set of maximal

$\mathbf{K}$ -consistent sets and $\Gamma R^c \Delta$ iff $\{A : \Box A \in \Gamma\} \subseteq \Delta$. A routine induction (the Truth Lemma) shows every consistent formula is satisfiable in $\mathfrak{M}^c$. $\square$

References

- [1] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Vol. 53. Cambridge Tracts in Theoretical Computer Science. Cambridge: Cambridge University Press, 2001.